Unit - IV

PROJECT MANAGEMENT USING NETWORK ANALYSIS

Introduction:

Network model is a powerful tool in the management of projects, particularly those consisting of large number of activities related in complex ways. Most realistic projects that organizations undertake are large and complex. Almost every industry worries about how to manage their large-scale and complicated projects effectively. Drawing a network model provides a visual display of these relationships concerned and a way of answering various questions about the project. Programme Evaluation and Review Techniques (PERT) and the Critical Path Method (CPM) are two popular techniques that help managers plan, schedule, monitor and control large and complex projects. Common application of network analysis occurs in project scheduling for:

i. Construction
ii. Engineering
iii. Manufacturing
iv. The management of administrative systems.

Methodology of CPM and PERT:

i. Define the project and all its significant activities.
ii. Develop the relationship among the activities by deciding which activities must proceed and follow others.
iii. Draw the network connecting all of the activities.
iv. Assign time and/or cost estimates of each activity.
v. Compute the longest time path through the network (this is called the critical path).
vi. Use the network to help plan, scheduled, monitor and control project.

Basic Terminology:

i. **Activity:** This is a task or job that consumes time or resources e.g. 'Verify debtors in a sales ledger'; 'Dig foundation'; 'Type report', etc. The symbol used is an arrow. The head of the arrow indicates where the job ends and the tail where the job starts.

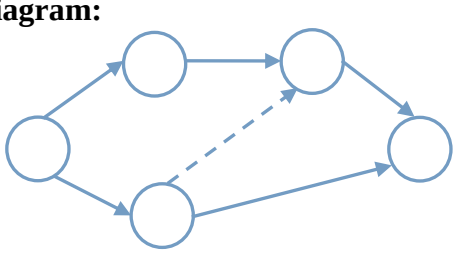
ii. **Dummy Activity:** This is an activity that does not require time or resources. It is used merely to show logical dependencies between activities so as not to violate the rules of drawing networks. The symbol used is an arrow with broken lines.

iii. **Event/Node:** It is a point in time that marks the start or end of an activity. The symbol used is a circle with a number to locate its position.



iv. **Network:** This is a combination of activities, dummy activities and events in a logical sequence according to the rules of drawing networks.

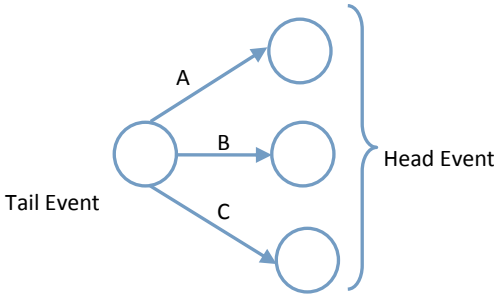
Example of a Network Diagram:



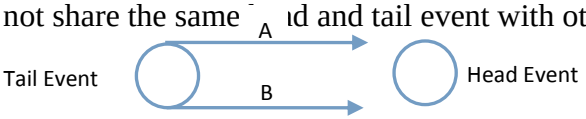
Rules for Drawing Network Diagrams:

i. A complete network should have only one point of entry (a start event) and only on point of exit (a finish event).

ii. Every activity must begin with one tail event (predecessors) and end with one head event (successor).

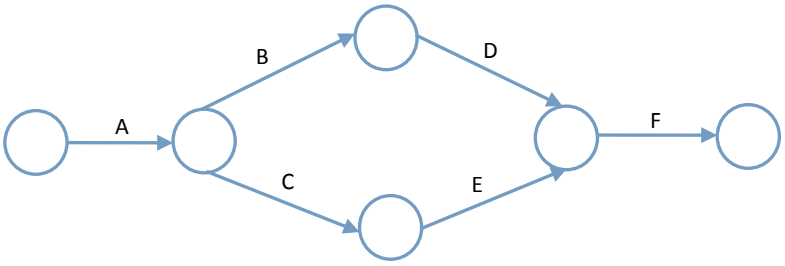


Note: An activity must not share the same head and tail event with other Activity.



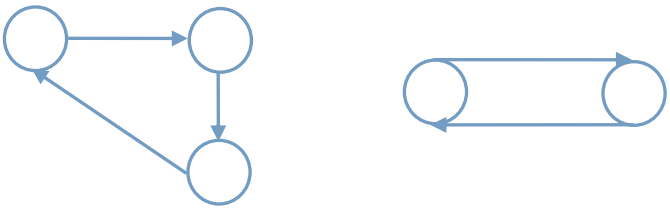
This is not allowed.

i. No activity can start until its tail event is reached.



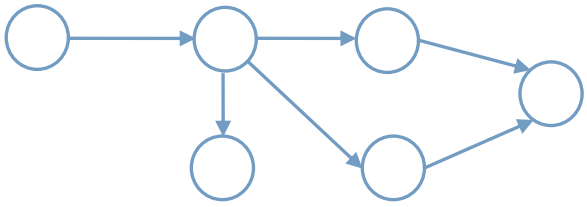
Event F cannot start until all activities (D and E) leading to it are completed.

ii. Loops are not allowed. Loops are activities leading to the same Event.



These are not allowed.

iii. Dangling activities are not allowed; in other words, all activities must contribute to the Network diagram otherwise they are discarded as irrelevant.



Dangling

iv. Network proceeds from left to right.

Note:

i. Network diagram are not drawn to scale as arrows does not represent time elapse.
ii. Arrows need not be drawn in the horizontal plane.
iii. Events should be progressively numbered from left to right e.g. 0 , 1 , 2 ,... or 0 , 10 , 20 , 30 ,...etc.
iv. Activities can be identified by their:
    a. Short description e.g. Type a report.
    b. Alphabetic or numeric code e.g. A , B , C ,... or 1 , 2 , 3 ,...
    c. Tail & head events numbers e.g. 1 - 2, 2 - 3, 3 - 5 etc
v. Use of Dummy Activities: Dummy activities are merely used to complete a network diagram, for a good logical relationship.

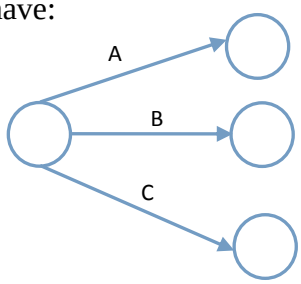
Problem 1: A project consists of the following activities as tabulated below:

Activity	Preceding Activity
A	-
B	-
C	-
D	A
E	A
F	B
G	B
H	C,F
I	D,G,H

Construct the network of the above project using activity-on-arrow Diagram.

Solution:

Stage 1: The first three activities A , B and C start the event of the network diagram because nothing precedes them. Thus, we have:



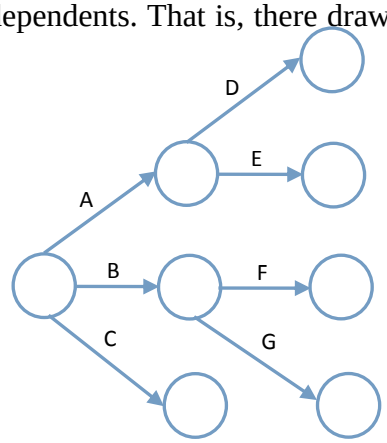
Note: The order of arrangement is not important. This may change in the course of during the complete diagram.

Stage 2

Case 1: Activities D and E preceded by activity A. or,

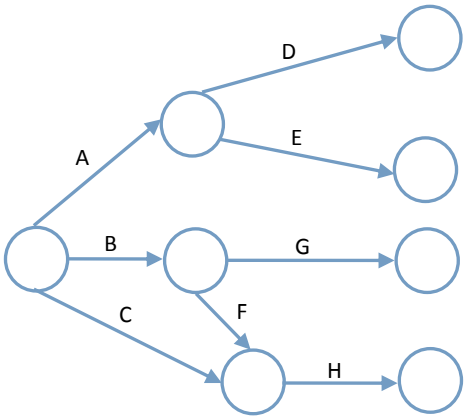
Case 2: Activities F and G are preceded by activity B.

These two events are independents. That is, there drawing does not affect one another. Thus, we have:

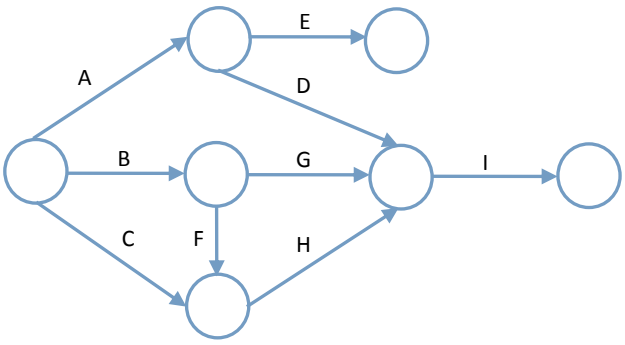


Note: Also the position of D and E can be interchanged depending on the relationships that may exist further.
Similarly, F and G .

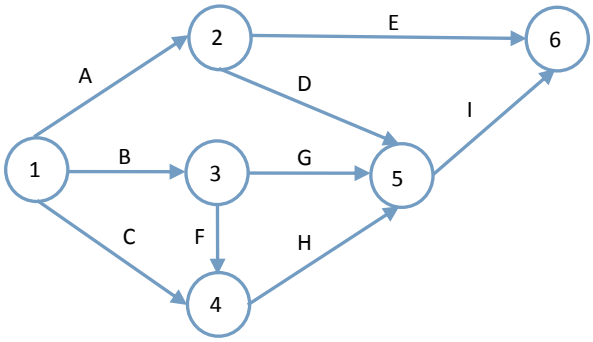
Stage 3: Activity H is preceded by activities C and F . This implies that C and F must end at the same point before H can start. This can be achieved in the diagram by drawing the arrows of C and F to meet at the same node, then the arrow of H then takes off. Note that positions of G and F are interchange for proper connection.



Stage 4: Activity I is preceded by activities D , G and H . This implies that D, G and H must end at the same point before I can start. This can be achieved drawing the arrows of D , G and H to meet at the same node, then the arrow of I is then drawn. Thus:



Note: Positions of D and E were interchanged for proper connection. Since dangling is not allowed in Network diagram, hence we have below a complete Network diagram to the problem.



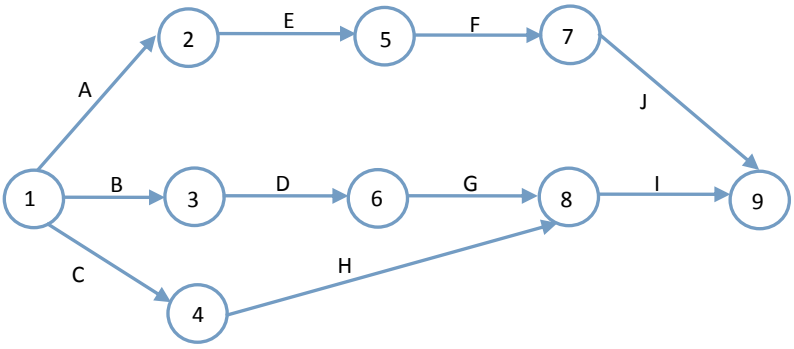
Note:

i. The events have been numbered 1 - 7 with 1 as start event & 7 as finish event.
ii. The shape of the diagram does not matter but the logical arrangement must be correct.
iii. Under examination condition, you are not expected to draw network diagram stage by stage as illustrated above; but a complete diagram is necessary. This is for illustration purpose. However, in drawing a network diagram, you will still have to go through those stages, but you need not break them down.

Problem 2: Activities necessary for test launch.

Activity	Description	Preceding Activity
A	Decide test market area	-
B	Agree marketing strategy	-
C	Agree production - specification	-
D	Decide brand name	B
E	Prepare advertising plan	A
F	Agree advertising package	E
G	Design packaging	D
H	Production of test batch	C
I	Package and distribute	G,H
J	Monitor media support	F

Construct the network of the above project using activity-on-arrow Diagram.



Time Analysis:

Critical Path/Time Determination (CPM Method): The critical path of a project or a network is the shortest time by which the whole project can be completed. In other words, it is the path of activities with the longest duration time. To determine the critical path, we use the following methods:

i. Calculating Earlier Start Time (EST) and Latest Start Time (LST) which employ the forward and backward pass rule respectively. However, to be sure, you need to check that the measure called total float is equal to zero (we shall deal with this in later section)
ii. Listing the possible paths in the network. The path with the longest duration is the critical path.

Calculation of Forward Pass Computations (to calculate Earliest, Time Ei):

Steps:

i. Begin from the start event and move towards the end event.
ii. Put E = 0 for the start event.
iii. Go to the next event (i.e node 2) if there is an incoming activity for event 2, add calculate Ei of previous event (i.e event 1) and activity time.

Note: If there are more than one incoming activities, calculate Ei for all incoming activities and take the maximum value. This value is the Ei for event 2.

iv. Repeat the same procedure from step 3 till the end event.

Calculation of Backward Pass Computations (to calculate Latest Time Li):

Steps:

i. Begin from end event and move towards the start event. Assume that the direction of arrows is reversed.
ii. Latest Time Li for the last event is the earliest time. Ei of the last event.
iii. Go to the next event, if there is an incoming activity, subtract the value of Li of previous event from the activity duration time. The arrived value is Li for that event. If there are more than one incoming activities, take the minimum Ei value.
iv. Repeat the same procedure from step 2 till the start event.

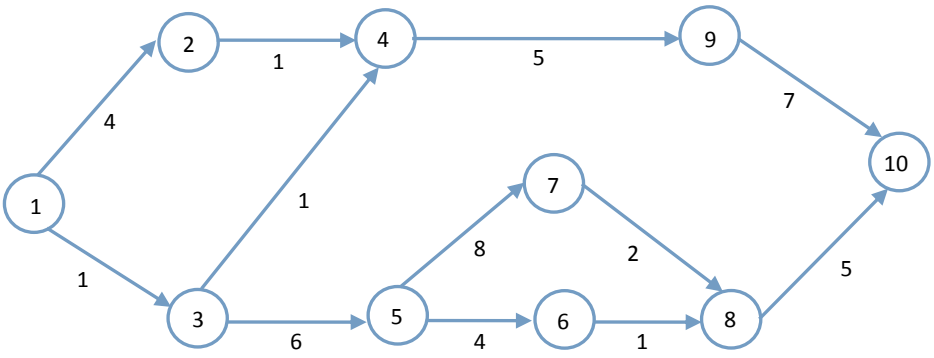
Problem 3: A project schedule has the following characteristics as shown in Table

Activity	Name	Time	Activity	Name	Time
1 - 2	A	4	5 - 6	G	4
1 - 3	B	1	5 - 7	H	8
2 - 4	C	1	6 - 8	I	1
3 - 4	D	1	7 - 8	J	2
3 - 5	E	6	8 - 10	K	5
4 - 9	F	5	9 - 10	L	7

i. Construct PERT network.
ii. Compute Ei and Li for each activity.
iii. Find the critical path.

Solution:

i. Construct PERT network:



ii. Compute Ei and Li for each activity

E_i values:

$E_1 = 0$; $E_2 = E_1 + t1, 2 = 0 + 4 = 4$

$E_3 = E_1 + t1, 3 = 0 + 1 = 1$;

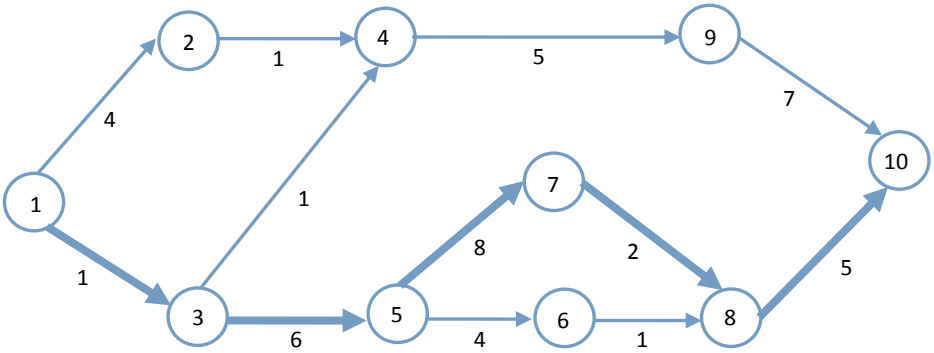
E_4 = max (E_2 + t2, 4 and E_3 + t3, 4)
= max (4 + 1 and 1 + 1) = max (5, 2)
= 5 days
E_5 = E_3 + t3, 6 = 1 + 6 = 7; E_6 = E_5 + t5, 6 = 7 + 4 = 11
E_7 = E_5 + t5, 7 = 7 + 8 = 15;
E_8 = max (E_6 + t6, 8 and E_7 + t7, 8)
= max (11 + 1 and 15 + 2) = max (12, 17)
= 17 days
E_9 = E_4 + t4, 9 = 5 + 5 = 10
E_{10} = max (E_9 + t9, 10 and E_8 + t8, 10)
= max (10 + 7 and 17 + 5) = max (17, 22)
= 22 days

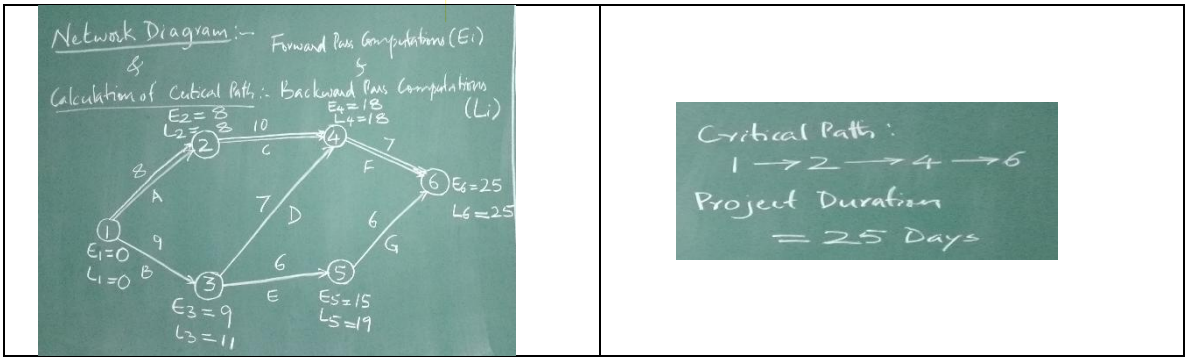
L_i values:

L_{10} = E_{10} = 22; L_9 = E_{10} – t9,10 = 22 – 7 = 15
L_8 = E_{10} – t8, 10 = 22 – 5 = 17; L_7 = E_8 – t7, 8 = 17 – 2 = 15
L_6 = E_8 – t6, 8 = 17 – 1 = 16;
L_5 = min (E_6 – t5, 6 and E_7 – t5, 7)
= min (16 – 4 and 15 –8) = min (12, 7)
= 7 days
L_4 = L_9 – t4, 9 = 15 – 5 =10
L_3 = min (L_4 – t3, 4 and L_5 – t3, 5)
= min (10 – 1 and 7 – 6) = min (9, 1)
= 1 day
L_2 = L_4 – t2, 4 = 10 – 1 = 9
L_1 = Min (L_2 – t1, 2 and L_3 – t1, 3)
= Min (9 – 4 and 1 – 1) = 0

Activity	Name	Time	Early Time		Late Time		Total Float
			Early Start	Early Finish	Late Start	Late Finish	
1 - 2	A	4	0	4	5	9	5
1 - 3	B	1	0	1	0	1	0
2 - 4	C	1	4	5	9	10	5
3 - 4	D	1	1	2	9	10	8
3 - 5	E	6	1	7	1	7	0
4 - 9	F	5	5	10	10	15	5
5 - 6	G	4	7	11	12	16	5
5 - 7	H	8	7	15	7	15	0
6 - 8	I	1	11	12	16	17	5
7 - 8	J	2	15	17	15	17	0
8 - 10	K	5	17	22	17	22	0
9 - 10	L	7	10	17	15	22	5

iii. From the Table , we observe that the activities 1 - 3, 3 - 5, 5 - 7,7 - 8 and 8 - 10 are critical activities as their floats are zero.





Float or Spare Time: Floats are spare time (idle time or unused time) available on non-critical activities. Activities on the critical paths always have zero (0) or no float and there are three types of float. They are:

i. Total float: This is the amount of time a path of activity could be delayed without affecting the overall project duration. The measure is as follows:

Total float = LFT - EST - D

ii. Free float: This is the amount of time an activity can be delayed without affecting the commencement of a subsequent activity at its Earliest Start Time, but may affect float of previous activity. The measure is as follows:

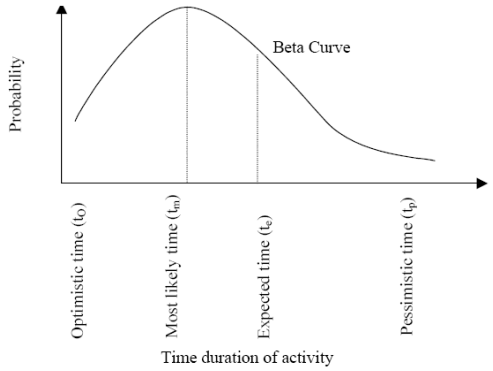
Free float = EFT - EST - D

iii. Independent float: This is the amount of time an activity can be delayed when all preceding activities are completed as late as possible and all succeeding activities are completed as early as possible. The measure is as follows:

Independent float = EFT - LST - D

Project Evaluation Review Technique (PERT):

In the critical path method, the time estimates are assumed to be known with certainty. In certain projects like research and development, new product introductions, it is difficult to estimate the time of various activities. Hence PERT is used in such projects with a probabilistic method using three time estimates for an activity, rather than a single estimate, as shown in Figure.



i. **Optimistic time T_o :** It is the shortest time taken to complete the activity. It means that if everything goes well then there is more chance of completing the activity within this time.

ii. **Most likely time T_m:** It is the normal time taken to complete an activity, if the activity were frequently repeated under the same conditions.

iii. **Pessimistic time T_p:** It is the longest time that an activity would take to complete. It is the worst time estimate that an activity would take if unexpected problems are faced.

Taking all these time estimates into consideration, the expected time of an activity is arrived at.

$$Te = (T_o + 4T_m + T_p) / 6$$

The variance of the activity time is calculated using the formula:

$$\sigma^2 = [(T_p - T_o)/6]^2$$

Probability for Project Duration: The probability of completing the project within the scheduled time (Ts) or contracted time may be obtained by using the standard normal deviate where Te is the expected time of project completion.

$$Z_0 = \frac{T_s - T_e}{\sqrt{\Sigma\sigma^2 \text{ in critical path}}}$$

Probability of completing the project within the scheduled time is

$$P(T < T_s) = P(Z < Z_0) \text{ from Normal Distribution tables}$$

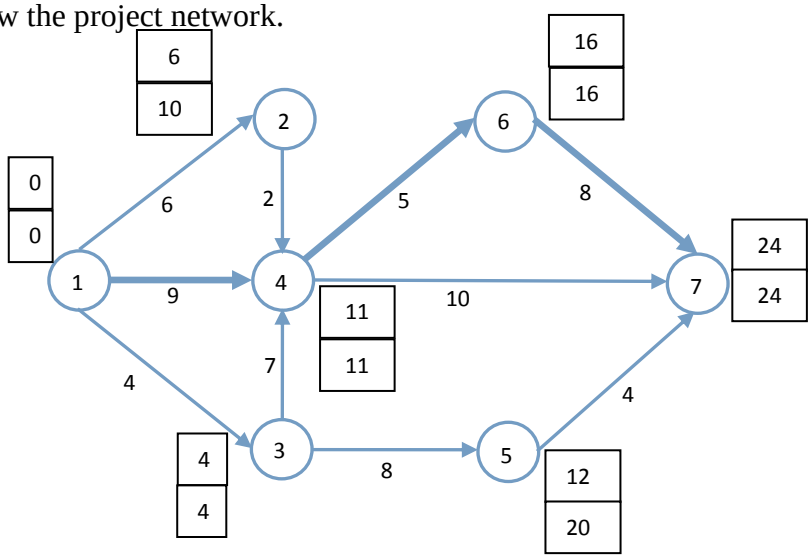
Problem 4: An R & D project has a list of tasks to be performed whose time estimates are given in the Table, as follows:

Activity	Activity Name	T_o	T_m	T_p
1 - 2	A	4	6	8
1 - 3	B	2	3	10
1 - 4	C	6	8	16
2 - 4	D	1	2	3
3 - 4	E	6	7	8
3 - 5	F	6	7	14
4 - 6	G	3	5	7
4 - 7	H	4	11	12
5 - 7	I	2	4	6
6 - 7	J	2	9	10

i. Draw the project network.
ii. Find the critical path.
iii. Find the probability that the project is completed in 19 days. If the probability is less than 20%, find the probability of completing it in 24 days.

Solution:

i. Draw the project network.



From the network diagram Figure, the critical path is identified as 1-4, 4-6, 6-7, with a project duration of 24 days.

Time expected for each activity is calculated using the formula:

$Te = (T_o+4T_m+T_p) / 6$

$= [4+4(6)+8] / 6$

$= 6$ days for activity A

The variance of activity time is calculated using the formula:

$\sigma^2 = [(T_p - T_o)/6]^2$

$= [(8 - 4)/6]^2$

$= 0.444$

Similarly, expected time and variances of all the activities are calculated as:

Activity	Activity Name	T_o	T_m	T_p	Te	σ^2
1 - 2	A	4	6	8	6	0.444
1 - 3	B	2	3	10	4	1.777
1 - 4	C	6	8	16	9	2.777
2 - 4	D	1	2	3	2	0.111
3 - 4	E	6	7	8	7	0.111
3 - 5	F	6	7	14	8	1.777
4 - 6	G	3	5	7	5	0.444
4 - 7	H	4	11	12	10	1.777
5 - 7	I	2	4	6	4	0.444
6 - 7	J	2	9	10	8	1.777

The probability of completing the project within 19 days is given by, $P (Z< Z_0)$

To find Z_0

$$Z_0 = \left(\frac{T_s - T_e}{\sqrt{\Sigma\sigma \text{ in critical path}}} \right)$$

$$= \left(\frac{19 - 22}{\sqrt{2.777 + 0.444 + 1.777}} \right)$$

$$=\left(\frac{-3}{\sqrt{5}}\right)=-1.3416$$

we know, P (Z <Z Network Model 0) = 0.5 – z (1.3416) (from normal tables, z (1.3416) = 0.4099)

= 0.5 – 0.4099
= 0.0901
= 9.01%

Thus, the probability of completing the R & D project in 19 days is 9.01%.

Since the probability of completing the project in 19 days is less than 20% As in question, we find the probability of completing it in 24 days.

$$Z_0=\frac{T_s-T_e}{\sqrt{\Sigma\sigma \text{ in critical path}}}$$

$$=\left(\frac{24-22}{\sqrt{5}}\right)=\left(\frac{2}{\sqrt{5}}\right)=0.8944 \text{ days}$$

$P(Z \leq Z_0)$ = 0.5 - Y (0.8944) from normal table Y (0.8944) = 0.3133

= 0.5 + 0.3133
= 0.8133
= 81.33%

Crashing of Network:

The two important components of any activity are the cost and time. Cost is directly proportional to time and vice versa.

i. Normal time: Normal time is the time required to complete the activity at normal conditions and cost.
ii. Crash time: Crash time is the shortest possible activity time; crashing more than the normal time will increase the direct cost.
iii. Cost Slope: Cost slope is the increase in cost per unit of time saved by crashing.

Procedure for Crashing:

Steps:

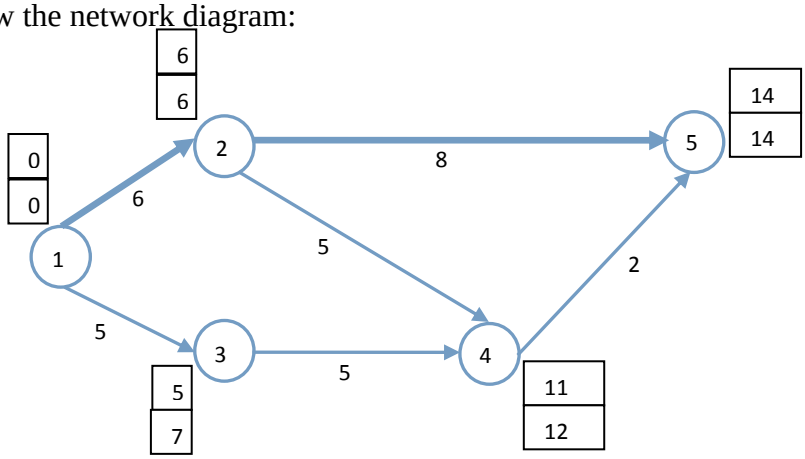
i. Draw the network diagram and mark the Normal time and Crash time.
ii. Calculate Ei and Li for all the activities.
iii. Find the critical path and other paths.
iv. Find the slope for all activities and rank them in ascending order.
v. Establish a tabular column with required field.
vi. Select the lowest ranked activity; check whether it is a critical activity. If so, crash the activity, else go to the next highest ranked activity.
Note: The critical path must remain critical while crashing.
vii. Calculate the total cost of project for each crashing
viii. Repeat Step 6 until all the activities in the critical path are fully crashed.

Problem 5: Determine the optimum project duration and cost for the following data:

Activity	Normal		Crash	
	Time (days) (N_T)	Cost (Rs) (N_c)	Time (days) (C_T)	Cost (Rs) (C_c)
1-2	6	50	4	80
1-3	5	80	3	150
2-4	5	60	2	90
2-5	8	100	6	300
3-4	5	140	2	200
4-5	2	60	1	80

If the indirect cost is Rs. 20 per day, crash the activities to find the minimum duration of the project and the project cost associated.

Solution: Draw the network diagram:



From the diagram, we observe that the critical path is 1-2-5 with project duration of 14 days

The cost slope for all activities and their rank is calculated as:

Cost slope = (Crash Cost - Normal Cost) / (Normal Time - Crast Time)

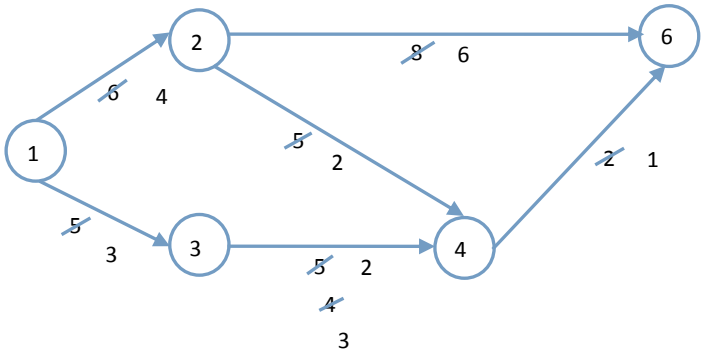
Cost slope for activity 1 - 2 = (80 - 50) / (6 - 4)

= 15

Activity	Cost Slope	Rank
1-2	15	2
1-3	35	4
2-4	10	1
2-5	100	5
3-4	20	3
4-5	20	3

The available paths of the network are listed down indicating the sequence of crashing.

Path	Number of days crashed
1 - 2 - 5	~~14~~ ~~12~~ ~~11~~ 10
1 - 2 - 4 - 5	~~13~~ ~~11~~ 10
1 - 3 - 4 - 5	~~12~~ ~~11~~ 10



The sequence of crashing and the total cost involved is given in below.

Initial direct cost = sum of all normal costs given

= Rs. 490.00

Activity Crashed	Project Duration	Critical Path	Direct Cost (RS.)	indirect Cost (RS.)	Total cost (Rs.)
-	14	1 - 2 - 5	490	14 x 20 = 280	770
1 - 2 (2) 2 - 5 (2) 2 - 4 (1) 3 - 4 (2)	10	1 - 2 - 5 1 - 3 - 4 - 5 1 - 2 - 4 - 5	490+(2x15)+(1x10)+(2x20) =770	10 x 20 = 200	970

It is not possible to crash more than 10 days, as all the activities in the critical path are fully crashed. Hence the minimum project duration is 10 days with the total cost of Rs. 970.00.